Stiffness of resistant breast-cancer cells from AFM force–distance curves

Frequentist, Bayesian and machine-learning analysis

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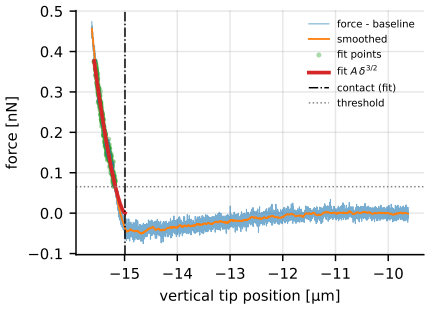
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Question and data

Question. Is the nucleus mechanically different from the cytoplasm in resistant breast-cancer cells, and does it differ between lines?

Data (JPK force files).

- 257 curves, 4 groups: MCF7R / T47DR × cytoplasm / nucleus (55, 60, 59, 83)
- cantilever MLCT-BIO; 1 session per line (25/02 and 18/03)
- extend segment → mechanics; retract segment → adhesion
- "good curves" pre-selected by the lab: no defective curves available



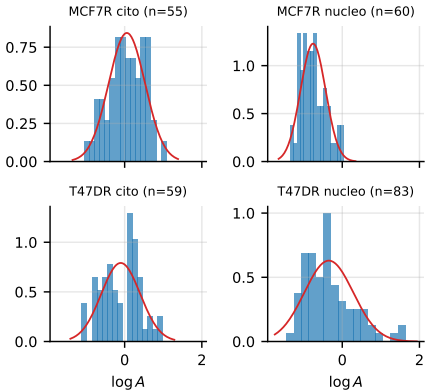
Baseline-corrected curve, contact point and Hertz-like fit.

From curves to one number

- robust linear baseline on the first 30% of the curve
- contact point from a threshold, then refitted
- Hertz-like model with free contact point:

$$F(\delta) = A\,\delta^{3/2}, \qquad \delta = z_c - z$$

- A = **effective** stiffness (nN μm$^{-3/2}$), *not* Young's modulus: tip radius and Poisson ratio are missing
- 9 preprocessing variants for systematics; median $R^2 = 0.986$
- the analysis works on $\log A$ (skewed A)



Pipeline: ten notebooks, one repository

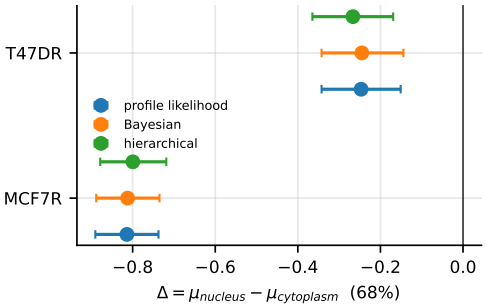
01	JPK reading, CSV export, diagnostic QC (nothing discarded)
02	baseline, contact point, Hertz-like fit, 9 variants
03	frequentist: profile likelihood, toy MC, Holm, systematics
04	Bayesian: MCMC, Bayes factors, priors, hierarchical, low statistics
05	ML: anomaly detection (injected defects), classification, mixtures
06	curve-shape ML: PCA, clustering, conformal prediction
07	depth-resolved stiffness, fixed-force check, T47DR shape signal
08	acquisition time, bursts of curves, temporal drift
09	calibration between sessions, repeatability, preconditioning
10	one Bayesian model: region, drift, position, burst

Notebooks 03–10 run from the tracked tables in `results/`; raw data are not in the repository.

First result: the nucleus is softer ($\Delta = \mu_{nucleus} - \mu_{cytoplasm}$ in $\log A$)

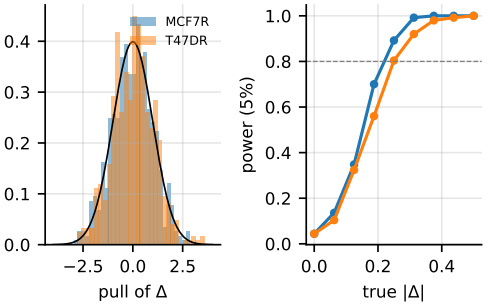
	MCF7R	T47DR
n (cyto / nuc)	55 / 60	59 / 83
Δ (68%)	$-0.81\,(-0.89, -0.74)$	$-0.25\,(-0.34, -0.15)$
ratio of A	0.44	0.78
p (Wilks)	$< 10^{-15}$	0.011
$\log_{10} BF_{10}$	13.9	0.6

- toy MC: unbiased, pull width 0.98–1.00, 68% coverage, size 5%
- 80% power needs $|\Delta| \gtrsim 0.22$–0.25: T47DR sits at the limit
- $p \simeq 0.01$ but Bayes factor only $\sim$ 4:1 for T47DR

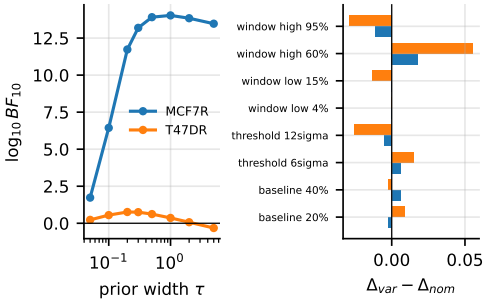


Three methods agree: profile likelihood, Bayesian, hierarchical.

Validation of the statistics



Toy MC: pulls and power of the Wilks test.

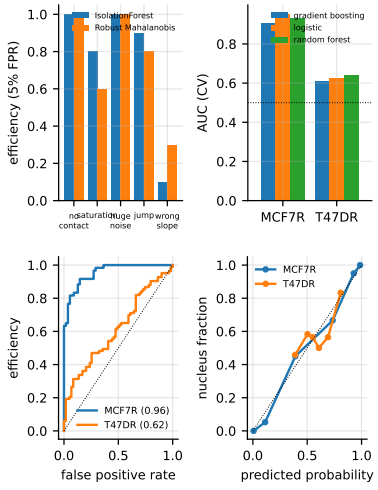


Bayes factor vs prior width; preprocessing systematics.

- systematics (9 variants, Barlow criterion) ≤ 0.07 in Δ, below the statistical error
- Student-t not preferred; posterior predictive check shows skewness in the nucleus groups

Machine learning

- **Anomaly detection** on 1000 synthetic curves with 5 injected defects: Isolation Forest / robust Mahalanobis, AUC 0.88 / 0.87; weak on `wrong_slope` (10–30%)
- **Region classifier:** AUC 0.95 (MCF7R), 0.63 (T47DR); transfers across lines
- **Curve shape:** in T47DR the shape alone gives AUC 0.76; it is the local exponent (nucleus 1.50 vs cytoplasm 1.43)
- **Conformal prediction:** T47DR sets are ambiguous for most curves

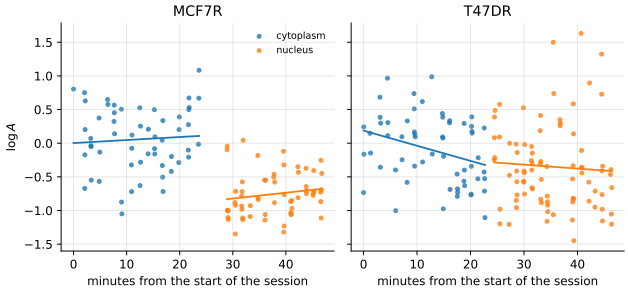


Stiffness without the Hertz fit

- **Fixed indentation:** $\Delta \log F(300\,\mathrm{nm}) = -0.76$ (MCF7R), -0.35 (T47DR)
- **Fixed force:** the nucleus needs a larger indentation: $\times 1.7$ (MCF7R), $+0.15$ to $+0.19$ in log (T47DR)
- the nucleus is softer at every depth between 50 and 600 nm
- free exponent ≈ 1.5: the model is adequate on average

But: these checks do not remove confounding from the experimental design ...

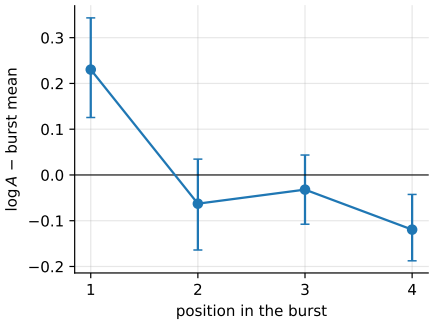
The design: regions measured in separate time blocks



- JPK timestamps: in both sessions cytoplasm first (0–23 min), nucleus after (25–47 min), *no overlap*
- region $\equiv$ time block
- T47DR: $\log A$ decreases during the session (-0.13 per 10 min); this drift alone predicts $\Delta \approx -0.30$
- MCF7R: drift goes the opposite way, so the effect grows (-0.97 ± 0.17)

Bursts, repeatability, preconditioning

- curves come in **bursts** of 3–5 at the same spot (2 s apart); 13–17 bursts per group
- intra-burst correlation moderate (ICC 0–0.18), design effect $\simeq 1.4$; SE of Δ +5% / +16%
- repeatability sd $= 0.33$–0.61 in $\log A$: scatter at one spot $\approx$ scatter between spots
- **first curve at a spot is stiffer**: second − first $= -0.25 \pm 0.09$ ($p = 0.013$)
- nucleus bursts are longer $\Rightarrow$ more softened curves in the nucleus mean
- first curve only: $\Delta = -0.77 \pm 0.13$ (MCF7R), -0.01 ± 0.26 (T47DR)



Calibration differs between sessions

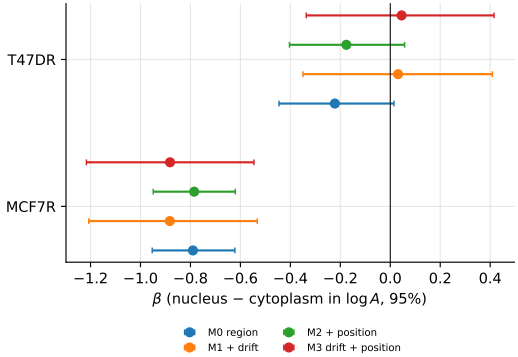
	25/02 (MCF7R)	18/03 (T47DR)
cantilever	2300FE	2300FE
spring constant	0.119 N/m	0.080 N/m
sensitivity	10.9 nm/V	14.0 nm/V
force setpoint	0.75 nN	0.75 nN

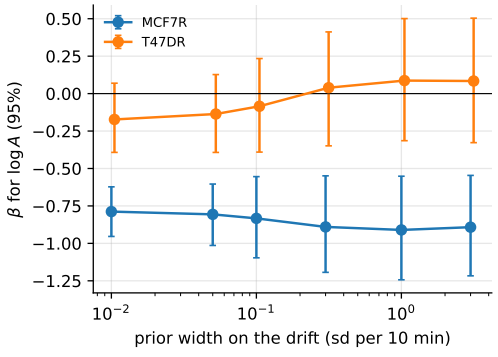
- same cantilever, -40% in k: shifts $\ln A$ by up to 0.4
- line differences (-0.16 cyto, $+0.41$ nucleus) cannot be interpreted, **not even in sign**
- inside a session k is common to both regions and cancels in Δ

One model for all of this

$$y_i = a + \beta\, r_i + \gamma\, t_i + \delta_{p_i} + u_b + \varepsilon_i, \qquad u_b \sim \mathcal{N}(0, \tau^2)$$

region r, time t (drift γ), position p in the burst, burst effect u_b integrated out; `emcee`





What survives

MCF7R: solid

- $\beta = -0.79$ without drift, -0.88 with drift and position; 95% $[-1.22, -0.55]$
- same for force at fixed indentation
- independent of the prior on the drift

T47DR: not established

- stiffness: $\beta = -0.22\,[-0.45, +0.02]$ with bursts; $+0.03\,[-0.35, +0.41]$ with drift
- $P(\beta < 0)$ from 0.92 (negligible drift) to 0.34 (free drift)
- **robust:** the nucleus has a higher contact-law exponent ($+0.062$, $[0.019, 0.103]$)

Effect difference between lines: $\beta_{MCF7R} - \beta_{T47DR} = -0.93\,[-1.42, -0.43]$ (calibration-free, still confounded with the session).

Limits and what would settle it

Limits

- region $\equiv$ time block; line $\equiv$ session; calibration differs by 40%
- no cell identifier (bursts as a proxy); pre-selected curves; effective parameter, not Young's modulus

Next measurements

- cytoplasm and nucleus **alternated** on the same cells, with the cell identifier
- calibration repeated within and between sessions
- first curve at each spot (or fewer indentations per spot)
- the "bad" curves, to validate anomaly detectors on real defects

Summary

- reproducible pipeline of 10 notebooks, results in a public repository (no raw data)
- nucleus softer than cytoplasm in MCF7R: **robust** to drift, bursts, preconditioning and the fit model
- T47DR: the stiffness difference is **not established**; a higher contact-law exponent in the nucleus is
- the experimental design (time blocks, calibration, preconditioning) limits what these data can say

`github.com/gmoretti03/AFM` · full report: `Docs/report/main.pdf`